ECA5601 –Advanced Wireless Communication Systems / 5G / 6G
Name: KV GAGAN YADAV
Reg no: 192412406

Ex.No.5: MIMO System Simulation with BER Analysis Date:12-8-26

AIM: To simulate a MIMO (Multiple Input Multiple Output) wireless communication system and analyze its performance in terms of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Channel Capacity (bps/Hz) using MATLAB.

Objective:

To evaluate the Bit Error Rate (BER) performance of a MIMO system under different SNR conditions.

Procedure:

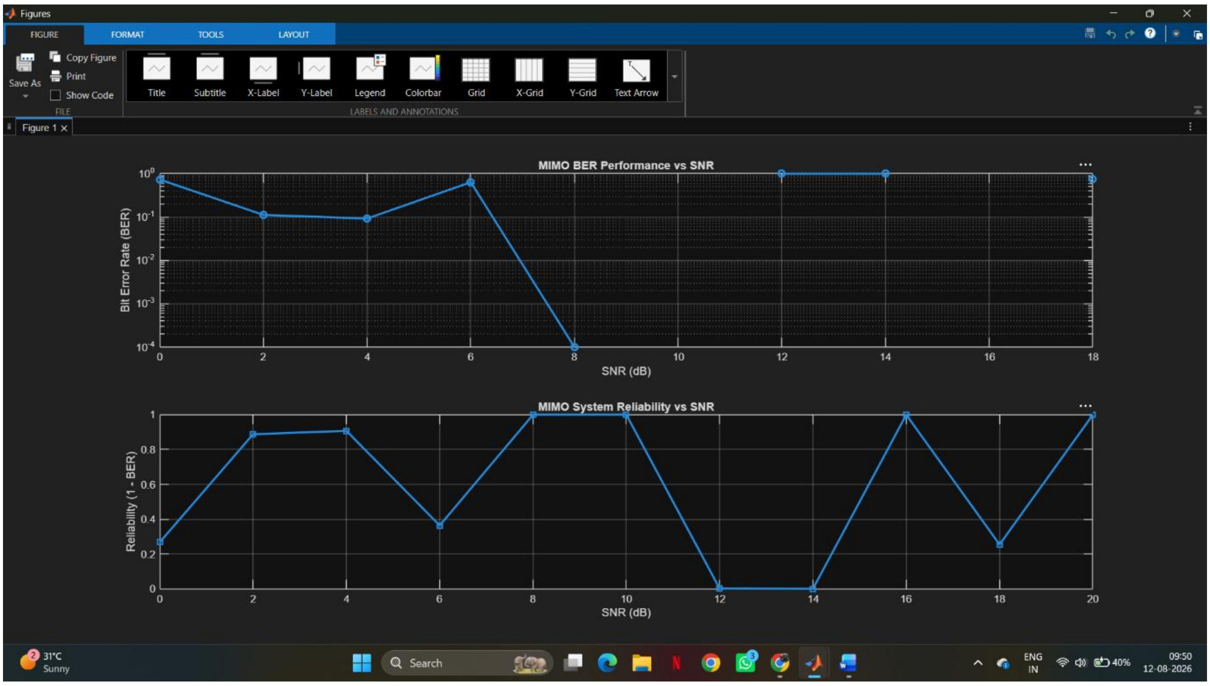
1. Define the MIMO system configuration (e.g., number of transmit and receive antennas such as 2×2 or 4×4).
2. Generate random binary data bits to represent the transmitted signal.
3. Modulate the bits using a digital modulation scheme (e.g., BPSK or QPSK).
4. Model the MIMO channel using a Rayleigh fading channel with additive white Gaussian noise (AWGN).
5. Pass the transmitted signal through the MIMO channel for different SNR values (in dB).
6. At the receiver, apply detection techniques (e.g., Zero-Forcing or Maximum Likelihood detection).
7. Demodulate the received signal to recover transmitted bits.
8. Compute Bit Error Rate (BER) by comparing transmitted and received bits.
9. Calculate Channel Capacity (bps/Hz) using SNR and MIMO capacity formula.
10. Plot BER vs SNR and Capacity vs SNR graphs for performance analysis.

Matlab Code :

```
clc; clear; close all;

% SNR range

SNR_dB = 0:2:20;
```

```
SNR = 10.^(SNR_dB/10);
% MIMO parameters
Nt = 2; Nr = 2;
BER = zeros(1,length(SNR));

for i = 1:length(SNR)
bits = randi([0 1],10000,1);
symbols = 2*bits-1; % BPSK
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));
tx = H(1,1)*symbols' + noise(1,:);
rx = real(tx) > 0;
BER(i) = sum(rx' ~= bits)/length(bits);
end
figure
% -------- Graph 1: BER vs SNR --------
subplot(2,1,1)
semilogy(SNR_dB,BER,'-o','LineWidth',2)
title('MIMO BER Performance vs SNR')
xlabel('SNR (dB)')
ylabel('Bit Error Rate (BER)')
grid on
% -------- Graph 2: SNR vs Quality Indicator --------
subplot(2,1,2)
plot(SNR_dB,1-BER,'-s','LineWidth',2)
title('MIMO System Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (1 - BER)')
grid on
```

Output:



Result:

1. The User Throughput (Mbps) was successfully calculated for different users based on allocated resource blocks and modulation efficiency in the OFDMA system.

2. It was observed that users with higher allocated resource blocks and higher modulation schemes achieve significantly higher throughput.

3. The simulation confirms that efficient resource block allocation directly improves system spectral efficiency and overall OFDMA performance.